

Data Science Project

Fraudulent Transaction Detection

Using sampling techniques to prevent false positive

Fraudulent credit card transactions are a global problem costing billions of dollars annually. Innovative procedures in classification methods allow for the detection of these transactions, ensuring stability in legitimate transactions and preventing excessive resource consumption by financial institutions. In this paper, using cutting-edge technology, three different machine learning techniques will be applied aiming for the highest level of accuracy, this being: logistic regression, gradient boosting, random forest and SMOTE for oversampling. The results shows that

Keywords — *Fraud detection; unbalanced database; smote;*

I. INTRODUCTION

Most payment fraud by value arose from credit transfers and card payments. More specifically, in 2024 the total value of fraudulent credit transfers sent by payment service providers (PSP2) in EU/EEA amounted to EUR 2.5 billion, with a fraud rate of 0.001%, and the value of fraudulent card transactions in the EU/EEA amounted to EUR 1.3 billion, with a rate of 0.033% [1]. In other words, fraud associated with credit, debit, and prepaid cards is a significant and growing issue for consumers, businesses, and the financial industry.

Despite the high values, the low fraud rate poses a challenge due to the non-stationary distribution of the data and the high imbalance in the class distribution. Likewise, public data is rare and commonly contains confidentiality issues, leaving open questions about the best approach [2].

This imbalance of the minority class favors the majority class, creating false high accuracy in detection algorithms even in cases where, in reality, performance is below expectations in the important class [3]. Nowadays, most Fraud Detection Systems (FDS) use increasingly sophisticated tools and algorithms with the aim of interpreting patterns and recognizing them as fraudulent both in real time and offline, always seeking to maintain the lowest rate of disturbances in genuine transactions. For example difficulties arise when learning from an unbalanced dataset as many machine intelligence methods are not designed to handle extremely large differences between class sizes [4].

Despite the existence of algorithms that can support large-scale unbalanced data, such as generative adversarial networks, adaptive graph-based representative sampling ensemble and adaptive synthetic sampling [X, Y, Z], research indicates that the use of the Synthetic Minority Over-Sampling Technique (SMOTE) can increase the success rate in detecting true-positive cases, since the use of regression mechanisms such as K-nearest neighbors does not assume that your data follows a specific linear or normal distribution, but rather that similar instances are close to each other, allowing the model to identify non-directly linear relationships, reducing the bias towards the majority class [3].

II. MATERIAL AND METHODS

A. Dataset

The dataset used in this paper can be obtained through the website Kaggle.com [5] and includes a subset of online European credit card transactions made in September 2013 over a period of two days.

Note that, due to the high level of imbalance in cases of legitimate transactions, represented by the number 0 in the class column, visible in Fig. 1, it is necessary to use sampling techniques in order to minimize negative effects during machine training.

	Count	Percentage (%)
Class		
0	284315	99.827251
1	492	0.172749

Fig 1. Class imbalance in the dataset

The low number of cases of fraudulent transactions, 492 cases, representing only 0.17% of the total number of registered transfers, can also generate incoherent results increasing the false negative rate. This low rate can also generate low recall, further hindering the language model's ability to recognize cases of fraud [2].

For confidentiality reasons, the dataset contains 28 columns named V1 to V28, the result of a PCA conversion, in addition to three other columns: Class, representing the legitimacy of the transfers; Time, representing the corresponding record time and Amount, which corresponds to the amount transferred during the transaction.

B. Methods

Imbalance learning: Imbalanced learning describes machine learning problems where one class (majority/negative) vastly outnumbers another (minority/positive), often by ratios of 100:1 or worse. Standard algorithms confronted with such data achieve artificially high accuracy by predicting everything as the majority class while completely ignoring rare minority events that typically carry the most business value. Most ML algorithms implicitly assume balanced class priors when optimizing objectives such as binary cross-entropy (BCE) loss:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log \log(\hat{y}_i) + (1 - y_i) \log \log(1 - \hat{y}_i) \right] \quad [6, \text{eq. (1)}]$$

When $y = \text{majority } 99\%$ of the time, minimizing L becomes trivial by predicting the majority class confidently. The few minority examples contribute negligibly to the total gradient signal.

Sampling: The sampling technique chosen for the tests was the *Synthetic Minority Over-Sampling Technique* (SMOTE), unlike techniques such as *Adaptive Synthetic Sampling* (ADASYN), which generates information based on data density, the SMOTE model generates this synthetic information based on interpolation between *K-nearest neighbors*, this guarantees the creation of uniform samples between the segments that join their neighbors [3].

Random Forest:

Logistic Regression:

Light GBM:

Validation:

III. RESULTS

A. Accuracy

B. Confusion Matrix

C. F1 Score

D. Area Under the Precision-Recall Curve (AUPRC)

IV. DISCUSSION

V. CONCLUSION

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